PID_Compact_1 [DB4]

PID_Compact_1 Propiedades							
General							
Nombre	PID_Compact_1	**Número**	4	**Tipo**	DB	**Idioma**	DB
Numeración	automática						
Información							
Título		**Autor**	SIMATIC	**Comentario**		**Familia**	
Versión	1.2	**ID personalizada**	PID_Cmpt				

Nombre	Tipo de datos	Valor de arranque	Remanencia	Accesible desde HMI	Visible en HMI	Valor de ajuste	Comentario
▼ Input							
Setpoint	Real	0.0	False	True	True	False	controller setpoint input
Input	Real	0.0	False	True	True	False	actual value of process as REAL
Input_PER	Word	W#16#0	False	True	True	False	actual value of process from periphery
ManualEnable	Bool	FALSE	False	True	True	False	activate manual input to overwrite output
ManualValue	Real	0.0	False	True	True	False	input for manual value
Reset	Bool	FALSE	False	True	True	False	reset the controller
▼ Output							
ScaledInput	Real	0.0	False	True	True	False	scaled peripheral input value from process
Output	Real	0.0	False	True	True	False	output value in REAL format
Output_PER	Word	W#16#0	False	True	True	False	output value in peripheral format
Output_PWM	Bool	FALSE	False	True	True	False	pulse width modulated output value
SetpointLimit_H	Bool	FALSE	False	True	True	False	setpoint is limited at highest level
SetpointLimit_L	Bool	FALSE	False	True	True	False	setpoint is limited at lowest level
InputWarning_H	Bool	FALSE	False	True	True	False	input value exceeded high warning level
InputWarning_L	Bool	FALSE	False	True	True	False	input value exceeded low warning level
State	Int	0	False	True	True	False	status of controller (0=Inactive,1=SUT, 2=TIR,3=Automatic,4=Hand)
Error	DWord	DW#16#00000000	False	True	True	False	error message from controller
InOut							
▼ Static							
sd_VersionID	DWord	DW#16#01020002	False	True	True	False	version of controller
sb_ResOld	Bool	FALSE	False	False	False	False	internal
sb_TMBeginExec	Bool	FALSE	False	False	False	False	internal
sb_GetCycleTime	Bool	TRUE	False	True	True	False	start automatic estimation of call cycle time
sb_EnCyclEstimation	Bool	TRUE	False	True	True	True	enable estimation of call cycle time
sb_EnCyclMonitoring	Bool	TRUE	False	True	True	True	enable monitoring of call cycle time
sb_Startup	Bool	FALSE	False	False	False	False	internal
sb_RunModeByStartup	Bool	FALSE	False	True	True	True	enable running last state before reset or startup
sby_EsData_1	Byte	16#01	False	False	False	True	internal
sby_EsData_2	Byte	16#01	False	False	False	True	internal
si_TMCnt	Int	0	False	False	False	False	internal
si_Unit	Int	0	False	True	True	True	unit of actual value
si_Type	Int	1	False	True	True	True	type of controller
si_SveModeByRes	Int	0	False	False	False	False	internal
sd_Warning	DWord	DW#16#00000000	False	True	True	False	warning message from controller
st_TMEnd	Time	T#0MS	False	False	False	False	internal
sr_TMDiff	Real	0.0	False	False	False	False	internal
sr_TMDiffMax	Real	0.0	False	False	False	False	internal
sr_TMDiffMaxMed	Real	0.0	False	False	False	False	internal
sr_TMDiffSum	Real	0.0	False	False	False	False	internal
▼ sBackUp	Struct		False	True	True	False	saved parameter set
r_Gain	Real	1.0	False	True	True	True	saved proportional gain
r_Ti	Real	20.0	False	True	True	True	saved integration time
r_Td	Real	0.0	False	True	True	True	saved derivative time
r_A	Real	0.0	False	True	True	True	saved filter coefficient for derivative part
r_B	Real	0.0	False	True	True	True	saved weigthing of proportional part in direct, feedback path
r_C	Real	0.0	False	True	True	True	saved weigthing of derivative part in direct, feedback path
r_Cycle	Real	1.0	False	True	True	True	saved cycle time of controller part
▼ sPid_Calc	Struct		False	True	True	False	self tuning data
r_Cycle	Real	0.1	False	True	True	True	call cycle time
r_Resol	Real	1.0	False	False	False	False	internal
b_CtrIn	Bool	FALSE	False	False	False	False	internal
b_Switch_On	Bool	FALSE	False	False	False	False	internal
b_Switch_Off	Bool	FALSE	False	False	False	False	internal
b_Jump_On	Bool	FALSE	False	False	False	False	internal
b_Jump_Off	Bool	FALSE	False	False	False	False	internal
b_Jump	Bool	FALSE	False	False	False	False	internal
b_CalcOn	Bool	FALSE	False	False	False	False	internal
b_SpCrossed	Bool	FALSE	False	False	False	False	internal
b_PvFilAdapt	Bool	FALSE	False	False	False	False	internal

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b_2StepDeadTm	Bool	FALSE	False	False	False	False	internal
b_LastPeriod	Bool	FALSE	False	False	False	False	internal
b_Part1	Bool	FALSE	False	False	False	False	internal
b_Part2	Bool	FALSE	False	False	False	False	internal
b_RunIn	Bool	FALSE	False	True	True	False	forced run in setpoint
b_FrstDerivRdy	Bool	FALSE	False	False	False	False	internal
b_SpOffLtd	Bool	FALSE	False	False	False	False	internal
b_TimeAdapt	Bool	FALSE	False	False	False	False	internal
b_CalcParamSUT	Bool	FALSE	False	True	True	False	recalculate parameters for startup tuning
b_CalcParamTIR	Bool	FALSE	False	True	True	False	recalculate parameters for tuning in run
i_CtrlTypeSUT	Int	1	False	True	True	True	tuning rule for SUT (0-CHR PID,1-CHR PI)
i_CtrlTypeTIR	Int	4	False	True	True	True	tuning rule for TIR (0-2-A PID auto,fast,slow;3-ZN PID;4-ZN PI;5-ZN P)
i_WPCyclMax	Int	0	False	False	False	False	internal
i_WPCycl	Int	0	False	False	False	False	internal
i_FilCyc	Int	0	False	False	False	False	internal
i_MaxPeriod	Int	0	False	False	False	False	internal
i_Ev4Step	Int	0	False	False	False	False	internal
i_RepeatProc	Int	0	False	False	False	False	internal
i_CntPeriod	Int	0	False	False	False	False	internal
i_CtrlValAdapt	Int	0	False	False	False	False	internal
i_Counter4	Int	0	False	False	False	False	internal
d_Counter1	DInt	0	False	False	False	False	internal
d_Counter2	DInt	0	False	False	False	False	internal
d_Counter3	DInt	0	False	False	False	False	internal
d_CycCounter	DInt	0	False	False	False	False	internal
d_CycCountEnd	DInt	0	False	False	False	False	internal
d_TOn	DInt	0	False	False	False	False	internal
d_TOff	DInt	0	False	False	False	False	internal
d_TSum	DInt	0	False	False	False	False	internal
d_TPnt1	DInt	0	False	False	False	False	internal
d_THys	DInt	0	False	False	False	False	internal
d_THysAlt	DInt	0	False	False	False	False	internal
r_Medi	Real	0.0	False	False	False	False	internal
r_Pv0	Real	0.0	False	False	False	False	internal
r_PvAlt	Real	0.0	False	False	False	False	internal
r_PvAlt2	Real	0.0	False	False	False	False	internal
r_PvAltSUT	Real	0.0	False	False	False	False	internal
r_PvMedi	Real	0.0	False	False	False	False	internal
r_LmnFilOld1	Real	0.0	False	False	False	False	internal
r_LmnFilOld2	Real	0.0	False	False	False	False	internal
r_Stabw_Pv_1	Real	0.0	False	False	False	False	internal
r_Stabw_Pv_2	Real	0.0	False	False	False	False	internal
r_SpAlt	Real	0.0	False	False	False	False	internal
r_Time	Real	0.0	False	False	False	False	internal
r_PvDxMax	Real	0.0	False	False	False	False	internal
r_TDxMax	Real	0.0	False	False	False	False	internal
r_Noise	Real	0.0	False	False	False	False	internal
r_Noise2	Real	0.0	False	False	False	False	internal
r_Dx0	Real	0.0	False	False	False	False	internal
r_Dx	Real	0.0	False	False	False	False	internal
r_Dx2	Real	0.0	False	False	False	False	internal
r_DxMax	Real	0.0	False	False	False	False	internal
r_DxMax2	Real	0.0	False	False	False	False	internal
r_DiffDx	Real	0.0	False	False	False	False	internal
r_DiffDx2	Real	0.0	False	False	False	False	internal
r_Break	Real	100.0	False	False	False	False	internal
r_BreakTm	Real	0.0	False	False	False	False	internal
r_X1	Real	0.0	False	False	False	False	internal
r_X2	Real	0.0	False	False	False	False	internal
r_Up	Real	0.0	False	False	False	False	internal
r_Down	Real	0.0	False	False	False	False	internal
r_Switch	Real	0.0	False	False	False	False	internal
r_AKrit	Real	0.0	False	False	False	False	internal
r_DKrit	Real	0.0	False	False	False	False	internal
r_LmnOpt	Real	0.0	False	False	False	False	internal
r_ErOld	Real	0.0	False	False	False	False	internal
r_TRelOld	Real	1.0	False	False	False	False	internal
r_StdAbw	Real	0.0	False	False	False	False	internal
r_Lmn0	Real	0.0	False	False	False	False	internal
r_LmnFacLim	Real	0.0	False	False	False	False	internal
r_Lmn1	Real	0.0	False	False	False	False	internal
r_Sp1	Real	0.0	False	False	False	False	internal
r_SpOff	Real	1.0	False	False	False	False	internal
r_SHys	Real	0.0	False	False	False	False	internal
r_Progress	Real	0.0	False	True	True	False	current progress in percent
▼ sPid_Cmpt	Struct		False	True	True	False	data of controling part
r_Sp_Hlm	Real	3.402822e+38	False	True	True	True	setpoint high limit

Nombre	Tipo de datos	Valor de arranque	Remanen-cia	Accesible desde HMI	Visible en HMI	Valor de ajuste	Comentario
r_Sp_Llm	Real	-3.402822e+38	False	True	True	True	setpoint low limit
r_Pv_Norm_IN_1	Real	0.0	False	True	True	True	low value (input range of scaling)
r_Pv_Norm_IN_2	Real	27648.0	False	True	True	True	high value (input range of scaling)
r_Pv_Norm_OUT_1	Real	0.0	False	True	True	True	low value (output range of scaling)
r_Pv_Norm_OUT_2	Real	100.0	False	True	True	True	high value (output range of scaling)
r_Lmn_Hlm	Real	100.0	False	True	True	True	output value high limit
r_Lmn_Llm	Real	0.0	False	True	True	True	output value low limit
b_Input_PER_On	Bool	FALSE	False	True	True	True	activate actual value of process from periphery
b_LoadBackUp	Bool	FALSE	False	True	True	False	restore last saved parameter record
b_InvCtrl	Bool	FALSE	False	True	True	True	activate control inversion
r_Lmn_Pwm_PPTm	Real	0.0	False	True	True	True	PWM minimum on time
r_Lmn_Pwm_PBTm	Real	0.0	False	True	True	True	PWM minimum off time
r_Man	Real	0.0	False	False	False	False	internal
b_ManOn	Bool	FALSE	False	False	False	False	internal
b_Sync	Bool	FALSE	False	False	False	False	internal
b_PIDInit	Bool	FALSE	False	False	False	False	internal
d_Per	DInt	0	False	False	False	False	internal
d_Per2	DInt	0	False	False	False	False	internal
r_PvSum	Real	0.0	False	False	False	False	internal
r_Pv_Hlm	Real	100.0	False	True	True	True	input value high limit
r_Pv_Llm	Real	0.0	False	True	True	True	input value low limit
r_Pv_HWrn	Real	40.0	False	True	True	True	input value warn level high
r_Pv_LWrn	Real	-3.402822E+38	False	True	True	True	input value warn level low
r_Ctrl_IntHlm	Real	3.402822e+38	False	False	False	False	internal
r_Ctrl_IntLlm	Real	-3.402822e+38	False	False	False	False	internal
r_Ctrl_IOutv	Real	0.0	False	True	True	False	signal of integral part
r_Ctrl_IRest	Real	0.0	False	False	False	False	internal
r_Ctrl_DRest	Real	0.0	False	False	False	False	internal
r_Ctrl_DRueck	Real	0.0	False	False	False	False	internal
d_Lmn_Pwm_Ptm	DInt	0	False	False	False	False	internal
r_Lmn_Pwm_Rest	Real	0.0	False	False	False	False	internal
r_Lmn_Outv	Real	0.0	False	False	False	False	internal
d_Lmn_Per_Outv	DInt	0	False	False	False	False	internal
▼ sParamCalc	Struct		False	True	True	False	data for recalculating parameters
b_SecndSetp	Bool	FALSE	False	False	False	False	internal
i_Event_SUT	Int	0	False	True	True	False	current phase of start up tuning
i_Event_TIR	Int	0	False	True	True	False	current phase of tuning in run
r_Tu	Real	0.0	False	False	False	False	internal
r_DxMaxSUT	Real	0.0	False	False	False	False	internal
r_Ks	Real	0.0	False	False	False	False	internal
r_TKrit	Real	0.0	False	False	False	False	internal
r_KKrit	Real	0.0	False	False	False	False	internal
r_LmnSUT	Real	0.0	False	False	False	False	internal
▼ sRet	Struct		True	True	True	False	retain data
b_EnableManOld	Bool	FALSE	True	False	False	False	internal
i_Mode	Int	0	True	True	True	True	set to change status (0=Inactive; 1=SUT;2=TIR;3=Automatic;4=Hand)
i_ModeOld	Int	0	True	False	False	False	internal
i_SveModeByEnMan	Int	0	True	False	False	False	internal
i_StateOld	Int	0	True	False	False	False	internal
r_Ctrl_Gain	Real	12.0	True	True	True	True	actual proportional gain
r_Ctrl_Ti	Real	1800.0	True	True	True	True	actual integration time
r_Ctrl_Td	Real	0.0	True	True	True	True	actual derivative time
r_Ctrl_A	Real	0.0	True	True	True	True	actual filter coefficient for derivative part
r_Ctrl_B	Real	1.0	True	True	True	True	actual weigthing of proportional part in direct, feedback path
r_Ctrl_C	Real	0.0	True	True	True	True	actual weigthing of derivative part in direct, feedback path
r_Ctrl_Cycle	Real	1.0	True	True	True	True	actual cycle time of controller part